Biomedical Equipment Troubleshooting Guide

Version 1.0

Purpose

This guide documents structured troubleshooting methods for Fresenius 2008T dialysis systems. It focuses on systematic diagnosis, technical documentation, and knowledge transfer. No patient information or proprietary service documentation is included.

Core Troubleshooting Workflow

1. Verify the reported problem.
2. Reproduce the fault safely.
3. Review alarms, self-tests, and debug values.
4. Inspect mechanical, electrical, hydraulic, and sensor systems.
5. Replace or repair only after confirming evidence.
6. Verify repair through calibration and complete functional testing.
7. Document findings and lessons learned.

Common Systems

Hydraulic System
Conductivity System
Temperature Control
Pressure System
Blood Leak Detection
Bibag System
Electronic Control Boards

Case Study: Machine 7367

Reported problem: Machine would not reach heat disinfection temperature beyond approximately 30°C.

Troubleshooting performed:
• Verified heater voltage.
• Replaced heater rod.
• Replaced thermistor assembly.
• Replaced triad.
• Swapped sensor board and power logic board with known-good boards.
• Completed pre/post temperature sensor calibration.
• Verified balancing chamber cable and related components.

Lesson learned: Complex failures may require systematic elimination of multiple subsystems and careful documentation when manufacturer guidance does not fully explain observed behavior.

Case Study: Machine 9011

Reported Bibag door open indication during dialysis. Verified actuator and Gen 2 board before replacing Bibag door. Machine passed self-tests after repair.

Documentation Best Practices

Record the complaint, observations, diagnostic data, repair actions, verification results, remaining concerns, and recommendations for future technicians.

Future Expansion

Add additional case studies, decision trees, photographs, flowcharts, and troubleshooting matrices as experience grows.